

TANMAY DEWANGAN

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Education

Georgia Institute of Technology

Dec 2028

Bachelor of Science in Computer Science (Intelligence and Cybersecurity Threads)

GPA: 3.84/4.0

- **Relevant Coursework:** Data Structures and Algorithms, Privacy Technology, Linear Algebra, Multivariable Calculus

Experience

AsthmaAware - Personalized Asthma Risk Forecasting

Jan 2026 – Present

Project Lead & Backend/ML Developer

Atlanta, GA

- Leading a **12-member student team** across frontend, backend, and machine learning to build a mobile app that helps asthma users identify environmental trigger risks through **interactive heatmaps and forecasts**.
- Built and integrated a **ConvLSTM-based forecasting model** to predict **spatiotemporal asthma risk factors** from pollution, weather, vegetation, and environmental raster data.
- Developed **FastAPI** backend integration logic to convert **grid-based model predictions** into **structured JSON responses** for React frontend heatmap visualization.

University of Arizona – CyVerse Lab

May 2024 – Dec 2025

Data Science & Research Intern

Tucson, AZ

- Standardized metadata across **270+ research projects** and optimized **7.2PB of datasets** by converting files into efficient formats, improving dataset discoverability and reducing query times within the open-source **CKAN** portal.
- Developed automated **data pipelines** in Python and an interactive Gradio UI enabling **12,000+ researchers** to publish and manage cloud-hosted data from **AWS S3, Google Cloud Storage**, and institutional repositories.
- Presented project outcomes with a live demo and Q&A at the iRODS Consortium Conference, driving adoption of new data workflows among **30+ researchers and institutional partners** and earning the Best Student Talk Award.

Seniors Empowered in Technology – Non-Profit Organization

Sep 2023 – Jul 2025

Founder & Software Developer

Phoenix, AZ

- Designed and taught **60+** weekly cybersecurity and digital-skills workshops across **3** senior centers.
- Authored **18** illustrated step-by-step guides reused by staff, driving **95%** learner satisfaction and **80%** repeat attendance
- Built a **Java (Spring Boot)** ticketing web application with **Gradle** to intake, categorize, and triage requests; processed **150+** tickets, implemented priority tracking, and cut median resolution from **5.2** to **1.8** days (**65%** faster).
- Built and led a **8-person** volunteer team, enabling reliable coverage and faster resident support.

Code Ninjas

Feb 2024 – Jul 2025

Coding Tutor

Phoenix, AZ

- Mentored **200+** students in block coding, Python, and JavaScript, guiding mini-projects from idea to demo.
- Introduced debugging checklists and rubrics, cutting evaluation time by **40%** and raising completion rates.

Projects

Georgia School Nutrition Equity Data Analysis | *Python, pandas, Tableau*

- Cleaned and merged **multi-source county-level datasets** for **all 159 Georgia counties**, including free/reduced lunches, SNAP, enrollment, lunch price, Farm-to-School, and rural-urban classification data.
- Conducted **correlation, group comparison, and outlier analyses**, then designed visualizations to communicate how socioeconomic and geographic factors relate to lunch affordability, nutrition transparency, and school meal access.

SnapSort - AI Photo Sorter | *Python, PyTorch, YOLOv8, OpenCV, dlib, GCP, Flask, Firebase*

- Built a Flask web app for photographers using **Python, OpenCV**, and **Pillow** to organize large photo sets by detected objects, faces, blur, lighting, orientation, and people count with Firebase user quotas and secure authentication.
- Engineered and deployed a **Windows File Explorer** extension to seamlessly integrate AI-driven photo organization into users' workflows, enhancing accessibility and adoption for both casual and professional photographers.

Technical Skills

Languages: Python, Java, JavaScript, HTML/CSS

Data/ML: pandas, NumPy, scikit-learn, PyTorch, OpenCV, YOLOv8, Jupyter

Backend/Web: Flask, FastAPI, REST APIs, Spring Boot, Gradio, React

Cloud/Tools: AWS S3, Google Cloud Storage, Cloud Run, Firebase, Git, Linux